MITHA DWI PRANINDYA

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SUMMARY

Applied Data Science undergraduate at Politeknik Elektronika Negeri Surabaya with experience in machine learning, computer vision, IoT analytics, and data visualization. Proven ability to develop AI-driven solutions, predictive models, and web-based analytics platforms across smart retail, energy monitoring, agriculture, and recommendation systems.

WORK EXPERIENCE

Data Scientist Internship - PT. PLN UP3 Surabaya Utara **Jan 2026 - May 2026**

Contributed to the development of a web-based anomaly detection system for electricity usage monitoring and fraud identification. Applied machine learning techniques to analyze abnormal consumption patterns and built real-time dashboards to support operational monitoring and data-driven decision making.

EDUCATION

Bachelor of Applied Data Sciene **August 2023 - Present**

Politeknik Elektronika Negeri Surabaya GPA 3.64/4.00

ACHIEVEMENTS

4th Runner-Up – Indonesian Flying Robot Contest (KRTI), Fixed Wing Division **Nov 2025**

Represented the Electronic Engineering Polytechnic Institute of Surabaya in the national Indonesian Flying Robot Contest (KRTI), a government-organized aerospace innovation program. Contributed as an official by managing coordination and ensuring compliance to support the fixed-wing UAV team's successful participation.

PROJECTS

PLN Anomaly Detection **Jan 2026 - May 2026**

- Developed a web-based anomaly detection system to monitor electricity usage, identifying irregular patterns and potential customer fraud.
- Applied machine learning models to detect abnormal consumption and generate actionable insights for operational teams.
- Built interactive dashboards for real-time monitoring, enabling data-driven decisions and improving response efficiency.

AVISTORE **Nov 2025**

Computer Vision Based Smart Retail Checkout System

- Designed and developed an AI-powered checkout solution utilizing YOLO object detection and TensorFlow deep learning models for automated product recognition.
- Enabled real-time product detection and classification from camera streams, reducing dependency on barcode scanning and manual cashier operations.
- Integrated automated transaction calculation and product management features into a web-based platform to enhance checkout speed and operational accuracy.

CAPSAI **Nov 2025**

Chili Analytics Platform Supported by IoT and Artificial Intelligence

- Designed and implemented an IoT-based smart monitoring system for chili cultivation, integrating environmental sensors and cloud analytics.
- Developed predictive models for yield estimation and disease detection using machine learning techniques to enhance agricultural productivity.
- Built an interactive data visualization dashboard to provide real-time insights, supporting precision agriculture and informed decision-making.

Research Project: Energy Monitoring and Battery Usage Analysis System for UAVs Using ESP32 **Nov 2025**

- Led the data analysis and system design for an IoT-based energy monitoring system for unmanned aerial vehicles (UAVs), focusing on real-time data acquisition and battery consumption analysis.
- Developed algorithms for predictive battery management and State of Charge (SOC) estimation, improving battery life prediction and operational efficiency of UAVs.
- Contributed to a data-driven approach for UAV mission planning, supporting energy-aware flight strategies for extended mission durations.

STYLEMATE **Feb 2025 - Jun 2025**

Website Mix and Match Outfit With Realtime Weather

- Integrated real-time weather data to generate context-aware clothing recommendations.
- Built a full-stack web application using Next.js and PostgreSQL for recommendation delivery and product management.
- Improved personalization by matching user preferences with clothing attributes and environmental conditions.

JobSense **May 2025 - Jun 2025**

AI-Powered Web Application for Personalized Job Recommendations

- Developed a web application that recommends job opportunities based on user skills, interests, and career preferences.
- Implemented machine learning techniques for personalized recommendation generation and job matching.
- Designed a user-friendly dashboard to visualize recommendation results and career insights.

ORGANIZATIONS

IEEE PENS – Head of Public Relations Division **Jan 2026 - Present**

- Managed the Public Relations division, handling communications and media outreach.
- Organized and promoted technical events, seminars, and workshops.

ESPYRO TEAM - KRTI ROBOTICS **Feb 2024 – Feb 2026**

- Contributed to the design, development, and testing of UAV systems for national competitions.
- Assisted in mission planning, documentation, and coordination for successful contest participation.

BEM PENS – Ministry of Social Affairs **Nov 2024 - Nov 2025**

- Planned and executed social programs for community empowerment and student welfare.
- Managed outreach initiatives and promoted active student participation.

CT Arsa Foundation Surabaya – Public Relations Officer **May 2024 – May 2025**

- Managed communications and public relations with partners and stakeholders.
- Organized community events and promoted social initiatives.

ADDITIONAL INFORMATION

Technical Skills

- Computer Vision: YOLO (v8/v11), TensorFlow, Image Processing, Object Detection, Object Tracking
- Machine Learning & AI: Regression, Classification, Clustering, Predictive Modeling, Feature Engineering
- Languages & Frameworks: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL (PostgreSQL, MySQL)
- Data Visualization: Tableau, Power BI, Grafana
- Tools & DevOps: Git, GitHub, Model Deployment basics

Soft Skills

- Core: Analytical Thinking, Complex Problem-Solving, Teamwork
- Management: Leadership, Project Coordination, Technical Communication

Certifications

- Advanced Data Visualization in Tableau (2024)
- Advanced Dashboard Design in Power BI (2024)
- Advanced Metrics & Monitoring in Grafana (2024)